World-Frame Algorithm for First-Order Inverse Dynamics Derivatives with Constraint Embedding

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***Abstract*—This supplement derives a world-frame variant of the first-order inverse dynamics derivative algorithm presented in the companion paper [1]. The body-frame Algorithm 1 of that paper contains an inner while loop that walks from each cluster to the root, applying a force transform at each step. Expressing all spatial quantities in the world frame can eliminate this inner loop: the off-diagonal derivative blocks for each cluster can then be formed collectively via matrix operations. This approach has been previously used for the special case of open chains [2]. For constraint-embedded systems, however, the cluster-to-cluster spatial force transform has additional structure (all forces from a child cluster accumulate onto a single output body in the parent) and this structure requires modification to the previously presented world-frame algorithm [2]. This document develops those modifications.**

I. Background

A. Notation

We roughly follow the companion papers [1] and [3], though with higher fidelity to [3] due to the emphasis on cluster semantics herein. We default to the nomenclature in the code to assist with its reading. By contrast, [3] more closely matches nomenclature used previously in the literature.

We assume that a kinematic connectivity graph with corresponding spanning tree $\mathcal{G}_\mathcal{S}$ and cluster tree $\mathcal{G}_\mathcal{C}$ has been constructed as in [3], and that the bodies have been numbered to satisfy the topological ordering constraints of [3] (Section III-B). We use k and ℓ for cluster indices, and i and j for body indices. We denote $\lambda(k)$ as the parent of cluster k in the cluster tree $\mathcal{G}_\mathcal{C}$. $\mathcal{C}_k$ is the set of body numbers in cluster k, with $n_{b,k} = |\mathcal{C}_k|$ and $\mathcal{C}_k(j)$ denoting the j-th body (ordered by body number). The output body $\Lambda(k) \in \mathcal{C}_{\lambda(k)}$ is the greatest spanning-tree ancestor in the parent cluster $\lambda(k)$ as shared by all bodies in cluster k; equivalently, $\Lambda(k) = \mathcal{C}_{\lambda(k)}(j^*)$, where $j^* = j^*(k)$ is the parent subindex giving the position of $\Lambda(k)$ within the parent cluster's body list. The output body of a cluster is called the parent body of an aggregate body (i.e., cluster) in [3].

We adopt the subtree notation of [2]: ν_k denotes the set of clusters in the subtree rooted at cluster k (including k), and $\overline{\nu}_k = \nu_k \setminus \{k\}$. When used as a block-row or block-column index, ν_k refers to all velocity DoFs of every cluster in ν_k. The velocity DoFs of cluster k alone are written as index $[k]$. Thus $\frac{\partial \boldsymbol{\tau}}{\partial \mathsf{q}}[\nu_k,\, \ell]$ denotes the rectangular block whose rows span the DoFs of ν_k and whose columns span the DoFs of cluster ℓ.

Sans-serif symbols (e.g., S_k, M_k) denote cluster-level quantities, which are block-stacked over the $n_{b,k}$ bodies of cluster k; upright bold symbols (e.g., $\mathbf{M}_i$) denote corresponding 6D body-level spatial vector quantities. A superscript 0 denotes a world-frame quantity, e.g., $\mathbf{S}_i^0$.

B. Prior results

As given in [2], the derivatives of inverse dynamics for an open-chain model can be expressed using world-frame quantities. For each body i, we consider

$$\mathbf{t}_1[i] = \breve{\mathbf{M}}_i^0\, \mathbf{S}_i^0, \tag{1}$$

$$\mathbf{t}_2[i] = \breve{\mathbf{B}}_i^0\, \mathbf{S}_i^0 + \breve{\mathbf{M}}_i^0\, \dot{\boldsymbol{\Upsilon}}_i^0, \tag{2}$$

$$\mathbf{t}_3[i] = \breve{\mathbf{B}}_i^0\, \dot{\boldsymbol{\Psi}}_i^0 + \breve{\mathbf{M}}_i^0\, \ddot{\boldsymbol{\Psi}}_i^0 + (\breve{\mathbf{f}}_i^0 \bar{\times}^*)\, \mathbf{S}_i^0, \tag{3}$$

$$\mathbf{t}_4[i] = (\breve{\mathbf{B}}_i^0)^\top \mathbf{S}_i^0. \tag{4}$$

Here, $\breve{\mathbf{f}}_i^0$, $\breve{\mathbf{M}}_i^0$, $\breve{\mathbf{B}}_i^0$ are composite (subtree-accumulated) world-frame forces, inertias, and Coriolis factors [2]; $\mathbf{S}_i^0$, $\dot{\boldsymbol{\Psi}}_i^0$, $\ddot{\boldsymbol{\Psi}}_i^0$ are the world-frame motion-subspace and associated kinematics matrices of body i; and $\dot{\boldsymbol{\Upsilon}}_i^0 = \dot{\mathbf{S}}_i^0 + \dot{\boldsymbol{\Psi}}_i^0$ combines the body-frame and parent-frame rates of the motion subspace (see [2] for additional description). The notation $\mathbf{t}_1[i]$ denotes the $6 \times n_i$ block of $\mathbf{t}_1$ corresponding to the i-th joint. The four derivative blocks are then:

$$\frac{\partial \boldsymbol{\tau}}{\partial \mathsf{q}}[\nu_i,\, i] = \mathbf{t}_4[\nu_i]^\top\, \dot{\boldsymbol{\Psi}}_i^0 + \mathbf{t}_1[\nu_i]^\top\, \ddot{\boldsymbol{\Psi}}_i^0, \tag{5}$$

$$\frac{\partial \boldsymbol{\tau}}{\partial \mathsf{q}}[i,\, \overline{\nu}_i] = (\mathbf{S}_i^0)^\top\, \mathbf{t}_3[\overline{\nu}_i], \tag{6}$$

$$\frac{\partial \boldsymbol{\tau}}{\partial \dot{\mathsf{q}}}[\nu_i,\, i] = \mathbf{t}_4[\nu_i]^\top\, \mathbf{S}_i^0 + \mathbf{t}_1[\nu_i]^\top\, \dot{\boldsymbol{\Upsilon}}_i^0, \tag{7}$$

$$\frac{\partial \boldsymbol{\tau}}{\partial \dot{\mathsf{q}}}[i,\, \overline{\nu}_i] = (\mathbf{S}_i^0)^\top\, \mathbf{t}_2[\overline{\nu}_i]. \tag{8}$$

These formulas do not directly generalize to constraint-embedded cluster trees. Every $\mathbf{t}_1[i]$, for example, has exactly 6 rows in the open-chain case, so stacking columnwise over a subtree yields a uniform $6 \times n_{\nu_i}$ matrix on which the products (5)–(8) are defined. For a cluster k with multiple bodies, the analogous block $\breve{\mathsf{M}}_k^0 \mathsf{S}_k^0$ has $6n_{b,k}$ rows, which can vary across clusters and breaks this uniform stacking. Section II introduces accumulation operations to resolve this issue.

C. World-Frame Quantities

For a cluster k with $n_{b,k}$ bodies, the full motion transform from the world frame to the cluster frames is the $6n_{b,k} \times 6$ matrix

$${}^{\mathcal{C}_k}\mathsf{X}_0 = \begin{bmatrix} {}^{\mathcal{C}_k(1)}\mathbf{X}_0 \\ \vdots \\ {}^{\mathcal{C}_k(n_{b,k})}\mathbf{X}_0 \end{bmatrix}, \tag{9}$$

where ${}^{\mathcal{C}_k(j)}\mathbf{X}_0 \in \mathbb{R}^{6\times 6}$ is the spatial motion transform from world frame to the local frame of body $\mathcal{C}_k(j)$.

In practice, world-frame quantities are computed per body by extracting the 6×6 block ${}^{\mathcal{C}_k(j)}\mathbf{X}_0$ from (9) for $j = 1, \ldots, n_{b,k}$ and applying the inverse transform to body-level quantities. For example, writing $i = \mathcal{C}_k(j)$ for the corresponding body:

$$\mathbf{S}^0_{k,i} = ({}^i\mathbf{X}_0)^{-1}\,\mathbf{S}_{k,i}, \tag{10}$$

$$\mathbf{v}^0_i = ({}^i\mathbf{X}_0)^{-1}\,\mathbf{v}_i, \tag{11}$$

$$\mathbf{f}^0_i = {}^i\mathbf{X}_0^\top\,\mathbf{f}_i, \tag{12}$$

$$\mathbf{M}^0_i = {}^i\mathbf{X}_0^\top\,\mathbf{M}_i\,{}^i\mathbf{X}_0. \tag{13}$$

Here $\mathbf{S}_{k,i} \in \mathbb{R}^{6\times n_k}$ denotes the 6-row block of the cluster motion subspace S_k corresponding to body i (distinct from the spanning-tree joint subspace of tree joint i alone), while $\mathbf{v}_i$, $\mathbf{f}_i$, $\mathbf{M}_i$ are the standard 6-vector or 6×6 spatial quantities (velocity, force, and inertia) for body i. The same block-extraction and world-frame transformation applies to $\dot{\mathbf{\Psi}}^0_{k,i}$, $\ddot{\mathbf{\Psi}}^0_{k,i}$, $\dot{\mathbf{\Upsilon}}^0_{k,i}$ (world-frame blocks of $\dot{\mathbf{\Psi}}_k$, $\ddot{\mathbf{\Psi}}_k$, $\dot{\mathbf{\Upsilon}}_k$). One other common quantity is required:

$$\mathbf{B}^0_i = (\mathbf{v}^0_i\times^*)\,\mathbf{M}^0_i - \mathbf{M}^0_i\,(\mathbf{v}^0_i\times) + (\mathbf{M}^0_i\,\mathbf{v}^0_i)\bar{\times}^*\,. \tag{14}$$

II. Removing Transforms in World-Frame Computations

The entire modification relative to [2] boils down to one main idea: a cluster with multiple bodies doesn't hand off a single 6-vector or 6×6 block when transformed to its parent – it hands off a sum over its bodies, landing in one slot (the output body) of the parent cluster. Everything below is the bookkeeping needed to make that summing precise in world-frame coordinates. For notational simplicity in what follows, we let $\ell = \lambda(k)$ and $j^* = j^*(k)$.

A. Inner products and the block-row sum

Off-diagonal derivative blocks require inter-cluster products of the form

$$\mathsf{S}_\ell^\top\,{}^k\mathsf{X}_\ell^\top\,\breve{\mathsf{M}}_k\,\mathsf{S}_k, \tag{15}$$

where ${}^k\mathsf{X}_\ell^\top$ transforms cluster-k force quantities into the local frames of cluster ℓ. Following [3] (Eq. (36)), ${}^k\mathsf{X}_\ell^\top$ has a single non-zero block row since each body in cluster k connects through the output body $\Lambda(k)$ at position j^* in cluster ℓ. This gives the structure

$$\mathsf{S}_\ell^\top\,{}^k\mathsf{X}_\ell^\top = \mathbf{S}^\top_{\ell,\Lambda(k)}\begin{bmatrix}{}^{\mathcal{C}_k(1)}\mathbf{X}^\top_{\Lambda(k)} & \cdots & {}^{\mathcal{C}_k(n_{b,k})}\mathbf{X}^\top_{\Lambda(k)}\end{bmatrix}, \tag{16}$$

When all quantities are expressed in the world frame, the 6×6 spatial transforms reduce to the identity.

As a result, the operation $\mathsf{S}_\ell^\top{}^k\mathsf{X}_\ell^\top\,\breve{\mathsf{M}}_k\,\mathsf{S}_k$ can be re-written:

$$\mathsf{S}_\ell^\top{}^k\mathsf{X}_\ell^\top\,\breve{\mathsf{M}}_k\,\mathsf{S}_k = \left(\mathbf{S}^0_{\ell,\Lambda(k)}\right)^\top\mathrm{BlockRowSum}\left(\breve{\mathsf{M}}^0_k\,\mathsf{S}^0_k\right). \tag{17}$$

where for any matrix $\mathsf{F} \in \mathbb{R}^{6n_{b,k}\times m}$ the operation

$$\mathrm{BlockRowSum}(\mathsf{F}) \;=\; \sum_{j=1}^{n_{b,k}} \mathsf{F}[j] \;\in\; \mathbb{R}^{6\times m} \tag{18}$$

sums the $6\times m$ blocks $\mathsf{F}[j]$ of F. This block-row sum captures the aggregation structure of the cluster-level force transform and allows the product to be computed without any transforms when all quantities are in the world frame.

B. Inertia propagation and the block-diagonal sum

Composite inertia propagates from cluster k to cluster ℓ via the congruence transform

$$\breve{\mathsf{M}}_\ell \;+=\; {}^k\mathsf{X}_\ell^\top\,\breve{\mathsf{M}}_k\,{}^k\mathsf{X}_\ell. \tag{19}$$

The sparsity of ${}^k\mathsf{X}_\ell^\top$ (only its j^*-th block row is nonzero) means (19) affects only the j^*-th diagonal block of $\breve{\mathsf{M}}_\ell$ according to

$$\breve{\mathsf{M}}_\ell[j^*] \;+=\sum_{i=1}^{n_{b,k}} {}^{\mathcal{C}_k(i)}\mathbf{X}^\top_{\Lambda(k)}\,\breve{\mathsf{M}}_k[i]\,{}^{\mathcal{C}_k(i)}\mathbf{X}_{\Lambda(k)}$$

When all spatial quantities are expressed in the world frame, the 6×6 transforms reduce to the identity. In this case, the congruence amounts to summing the $n_{b,k}$ diagonal 6×6 blocks of $\breve{\mathsf{M}}^0_k$ into the appropriate slot:

$$\breve{\mathsf{M}}^0_\ell[j^*] \;+=\; \mathrm{BlockDiagSum}(\breve{\mathsf{M}}^0_k), \tag{20}$$

where for any matrix $\mathsf{M} \in \mathbb{R}^{6n_{b,k}\times 6n_{b,k}}$ the operation

$$\mathrm{BlockDiagSum}(\mathsf{M}) \;=\; \sum_{j=1}^{n_{b,k}} \mathsf{M}[j] \;\in\; \mathbb{R}^{6\times 6}. \tag{21}$$

with $\mathsf{M}[j]$ denoting the j-th 6×6 diagonal block of M. Note that for open-chain systems, both the $\mathrm{BlockRowSum}$ and $\mathrm{BlockDiagSum}$ reduce to identity operations.

III. Updated Derivative Blocks

A. Per-body temporaries and system-level accumulators

Mirroring the temporaries in (1)-(4), for each body $i \in \mathcal{C}_k$, define the four $6\times n_k$ matrices

$$\mathbf{F}_{1,i} = \breve{\mathbf{M}}^0_i\,\mathbf{S}^0_{k,i}, \tag{22}$$

$$\mathbf{F}_{2,i} = \breve{\mathbf{B}}^0_i\,\mathbf{S}^0_{k,i} + \breve{\mathbf{M}}^0_i\,\dot{\mathbf{\Upsilon}}^0_{k,i}, \tag{23}$$

$$\mathbf{F}_{3,i} = \breve{\mathbf{B}}^0_i\,\dot{\mathbf{\Psi}}^0_{k,i} + \breve{\mathbf{M}}^0_i\,\ddot{\mathbf{\Psi}}^0_{k,i} + (\breve{\mathbf{f}}^0_i\bar{\times}^*)\,\mathbf{S}^0_{k,i}, \tag{24}$$

$$\mathbf{F}_{4,i} = (\breve{\mathbf{B}}^0_i)^\top\mathbf{S}^0_{k,i}, \tag{25}$$

Because all quantities share the world frame, no transform is needed when summing $\mathbf{F}_{m,i}$ contributions across bodies. Four system-level accumulator matrices $\mathbf{F}_m \in \mathbb{R}^{6\times n}$, $m = 1, 2, 3, 4$, are then stored with block-column entries for cluster k as:

$$\mathbf{F}_m[k] \;=\; \sum_{i\in\mathcal{C}_k}\mathbf{F}_{m,i} \tag{26}$$

B. Diagonal blocks

Considering equations (62) and (63) of [1], the diagonal blocks for cluster k are computed using world frame quantities

TABLE I: Body-frame vs. world-frame algorithm structure

Feature	Body frame (Alg. 1 of [1])	World frame (Alg. 1 Herein)
Temporaries	$\mathbf{t}_1$–$\mathbf{t}_4$ (body coords)	$\mathbf{F}_{1,i}$–$\mathbf{F}_{4,i}$ (world coords)
Inner while loop	Yes, ${}^{\ell}\mathsf{X}_{\lambda(\ell)}^{\top}$ per step	Eliminated via accumulation in temporaries
Off-diag. computations per cluster	One pair per ancestor	One pair, using ν_k columns
Inertia prop.	${}^{k}\mathsf{X}_{\lambda(k)}^{\top}\breve{\mathsf{M}}_k{}^{k}\mathsf{X}_{\lambda(k)}$	BlockDiagSum into $[j^*]$ slot

as:

$$\frac{\partial \boldsymbol{\tau}}{\partial \mathsf{q}}[k,k] = \sum_{i\in\mathcal{C}_k}\Big(\mathbf{F}_{1,i}^{\top}\ddot{\boldsymbol{\Psi}}_{k,i}^{0} + \mathbf{F}_{4,i}^{\top}\dot{\boldsymbol{\Psi}}_{k,i}^{0}\Big) + (\nabla_{\mathsf{q}_k}\mathsf{S}_k^{\top}\mathsf{F})\big|_{\mathsf{F}=\mathsf{f}_k}, \tag{27}$$

$$\frac{\partial \boldsymbol{\tau}}{\partial \dot{\mathsf{q}}}[k,k] = \sum_{i\in\mathcal{C}_k}\Big(\mathbf{F}_{1,i}^{\top}\dot{\boldsymbol{\Upsilon}}_{k,i}^{0} + \mathbf{F}_{4,i}^{\top}\mathbf{S}_{k,i}^{0}\Big). \tag{28}$$

The last term in (27) is nonzero only for clusters with configuration-dependent S_k.

C. Off-diagonal blocks with the parent

We next express equations (61), (62), (64), and (65) of [1] using world-frame quantities, while employing accumulated quantities to address the removal of inter-cluster transforms.

Let $\ell = \lambda(k)$ and $i = \Lambda(k)$. After accumulating (26), the four off-diagonal blocks are:

$$\frac{\partial \boldsymbol{\tau}}{\partial \mathsf{q}}\big[\nu_k,\ \ell\big] = \mathbf{F}_1[\nu_k]^{\top}\ddot{\boldsymbol{\Psi}}_{\ell,i}^{0} + \mathbf{F}_4[\nu_k]^{\top}\dot{\boldsymbol{\Psi}}_{\ell,i}^{0}, \tag{29}$$

$$\frac{\partial \boldsymbol{\tau}}{\partial \mathsf{q}}\big[\ell,\ \nu_k\big] = \big(\mathbf{S}_{\ell,i}^{0}\big)^{\top}\mathbf{F}_3[\nu_k], \tag{30}$$

$$\frac{\partial \boldsymbol{\tau}}{\partial \dot{\mathsf{q}}}\big[\nu_k,\ \ell\big] = \mathbf{F}_1[\nu_k]^{\top}\dot{\boldsymbol{\Upsilon}}_{\ell,i}^{0} + \mathbf{F}_4[\nu_k]^{\top}\mathbf{S}_{\ell,i}^{0}, \tag{31}$$

$$\frac{\partial \boldsymbol{\tau}}{\partial \dot{\mathsf{q}}}\big[\ell,\ \nu_k\big] = \big(\mathbf{S}_{\ell,i}^{0}\big)^{\top}\mathbf{F}_2[\nu_k], \tag{32}$$

These revisions are collected into the world-frame algorithm in Algorithm 1.

IV. Comparison with the Body-Frame Algorithm

Table I summarizes the key structural differences. The $\approx 33\%$ runtime reduction reported in the companion paper comes from eliminating the repeated transforms and replacing the inner while loop with a single pair of matrix products per cluster. For large clusters the BlockDiagSum/BlockRowSum propagation adds $O(n_{b,k})$ floating-point additions per cluster, which is modest compared to the prior cost of repeated transforms that increased in number with tree depth.

Algorithm 1 World-Frame First-Order Derivatives of ID

Require: q, $\dot{\mathsf{q}}$, $\ddot{\mathsf{q}}$, model

$\mathbf{v}_0 = \mathbf{0}$; $\mathbf{a}_0 = -\mathbf{a}_g$; $\mathbf{F}_m = \mathbf{0}_{6\times n}$ for $m = 1,2,3,4$
for $k = 1$ **to** N **do**
  $\mathsf{v}_k,\ \mathsf{a}_k,\ \dot{\boldsymbol{\psi}}_k,\ \ddot{\boldsymbol{\psi}}_k,\ \dot{\boldsymbol{\Upsilon}}_k$
  ${}^{\mathcal{C}_k}\mathsf{X}_0$ via recursive kinematics.
  for each $i \in \mathcal{C}_k$, using block ${}^{i}\mathbf{X}_0$ from ${}^{\mathcal{C}_k}\mathsf{X}_0$ **do**
    $\mathbf{v}_i^0,\ \mathbf{S}_{k,i}^0,\ \dot{\boldsymbol{\Psi}}_{k,i}^0,\ \ddot{\boldsymbol{\Psi}}_{k,i}^0,\ \dot{\boldsymbol{\Upsilon}}_{k,i}^0 \leftarrow ({}^{i}\mathbf{X}_0)^{-1}(\cdot)$
    $\mathbf{f}_i^0 \leftarrow {}^{i}\mathbf{X}_0^{\top}\mathbf{f}_i$
    $\breve{\mathbf{M}}_i^0 \leftarrow$ (13); $\breve{\mathbf{B}}_i^0 \leftarrow$ (14) using $\breve{\mathbf{M}}_i^0$, $\mathbf{v}_i^0$
    $\breve{\mathbf{f}}_i^0 = \mathbf{f}_i^0$
  end for
end for
for $k = N$ **to** 1 **do**
  for each $i \in \mathcal{C}_k$ **do**
    $\mathbf{F}_{1,i} = \breve{\mathbf{M}}_i^0\mathbf{S}_{k,i}^0$; $\mathbf{F}_{4,i} = (\breve{\mathbf{B}}_i^0)^{\top}\mathbf{S}_{k,i}^0$
    $\mathbf{F}_{2,i} = \breve{\mathbf{B}}_i^0\mathbf{S}_{k,i}^0 + \breve{\mathbf{M}}_i^0\dot{\boldsymbol{\Upsilon}}_{k,i}^0$
    $\mathbf{F}_{3,i} = \breve{\mathbf{B}}_i^0\dot{\boldsymbol{\Psi}}_{k,i}^0 + \breve{\mathbf{M}}_i^0\ddot{\boldsymbol{\Psi}}_{k,i}^0 + (\breve{\mathbf{f}}_i^0\bar{\times}^*)\mathbf{S}_{k,i}^0$
    $\frac{\partial \boldsymbol{\tau}}{\partial \mathsf{q}}[k,k]$ += $\mathbf{F}_{1,i}^{\top}\ddot{\boldsymbol{\Psi}}_{k,i}^0 + \mathbf{F}_{4,i}^{\top}\dot{\boldsymbol{\Psi}}_{k,i}^0$
    $\frac{\partial \boldsymbol{\tau}}{\partial \dot{\mathsf{q}}}[k,k]$ += $\mathbf{F}_{1,i}^{\top}\dot{\boldsymbol{\Upsilon}}_{k,i}^0 + \mathbf{F}_{4,i}^{\top}\mathbf{S}_{k,i}^0$
    $\mathbf{F}_m[k]$ += $\mathbf{F}_{m,i}$ for $m = 1,2,3,4$
  end for
  $\frac{\partial \boldsymbol{\tau}}{\partial \mathsf{q}}[k,k]$ += $\nabla_{\mathsf{q}_k}\mathsf{S}_k^{\top}\mathsf{F}\big|_{\mathsf{F}=\mathsf{f}_k}$
  if $\lambda(k) > 0$ **then**
    $\ell = \lambda(k)$; $i = \Lambda(k)$; $j^* = j^*(k)$
    $\frac{\partial \boldsymbol{\tau}}{\partial \mathsf{q}}[\nu_k,\ell] = \mathbf{F}_1[\nu_k]^{\top}\ddot{\boldsymbol{\Psi}}_{\ell,i}^0 + \mathbf{F}_4[\nu_k]^{\top}\dot{\boldsymbol{\Psi}}_{\ell,i}^0$
    $\frac{\partial \boldsymbol{\tau}}{\partial \mathsf{q}}[\ell,\nu_k] = (\mathbf{S}_{\ell,i}^0)^{\top}\mathbf{F}_3[\nu_k]$
    $\frac{\partial \boldsymbol{\tau}}{\partial \dot{\mathsf{q}}}[\nu_k,\ell] = \mathbf{F}_1[\nu_k]^{\top}\dot{\boldsymbol{\Upsilon}}_{\ell,i}^0 + \mathbf{F}_4[\nu_k]^{\top}\mathbf{S}_{\ell,i}^0$
    $\frac{\partial \boldsymbol{\tau}}{\partial \dot{\mathsf{q}}}[\ell,\nu_k] = (\mathbf{S}_{\ell,i}^0)^{\top}\mathbf{F}_2[\nu_k]$
    $\breve{\mathsf{M}}_{\ell}^0[j^*]$ += BlockDiagSum($\breve{\mathsf{M}}_k^0$)
    $\breve{\mathsf{B}}_{\ell}^0[j^*]$ += BlockDiagSum($\breve{\mathsf{B}}_k^0$);
    $\breve{\mathbf{f}}_{\ell}^0[j^*]$ += BlockRowSum($\breve{\mathbf{f}}_k^0$)
    f_ℓ += ${}^{k}\mathsf{X}_{\lambda(k)}^{\top}\mathsf{f}_k$ [body-frame force, for $\nabla\mathsf{S}$ term]
  end if
end for
return $\frac{\partial \boldsymbol{\tau}}{\partial \mathsf{q}}$, $\frac{\partial \boldsymbol{\tau}}{\partial \dot{\mathsf{q}}}$

References

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